# Kinam Kim

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## Research Interest

My research is on **world models for Physical AI**: making **video generative models** into simulators of the physical world [U1, C1-C3, P1], and turning those simulators into robot behavior [U2-U5, C4-C6, O1]. I work on sampling, conditioning and spatiotemporal consistency on the generative side, and on latent reasoning inside **VLA** policies, reinforcement learning and zero-shot sim-to-real on the robot side.

**Keywords:** World Models, Video Diffusion Models, Vision-Language-Action (VLA) Models, Sim-to-Real

## Honors and Service

- **Outstanding Paper Award**, Robotics: Science and Systems (RSS) 2026, for *FlashSAC* [C5]
- **Reviewer**, ICLR 2027, NeurIPS 2026, CoRL 2026

## Education

**KAIST, Graduate School of AI** — Daejeon, South Korea
*M.S./Ph.D. Integrated (Advisor: Prof. Jaegul Choo)* — *Feb. 2025 - Present*
- Expected graduation: February 2029 GPA: 4.0 / 4.3

**KAIST, School of Computing** — Daejeon, South Korea
*B.S. in Computer Science* — *Mar. 2019 - Feb. 2024*
- GPA: 3.9 / 4.3

## Work Experience

**NVIDIA**
*Research Intern* — *Jul. 2026 - Present*
- Latent-reasoning VLA with directly supervised reasoning states, under review [U2]

**Microsoft Research Asia** — Tokyo, Japan
*Research Intern* — *Feb. 2026 - Jul. 2026*
- Object-centric residual RL for VLA policies, zero-shot sim-to-real transfer (CoRL 2026) [C4]
- Residual RL transferred through a frozen VLA's own latents, under review [U4]

**KAIST DAVIAN Lab** — Daejeon, South Korea
*Undergraduate Research Intern* — *Aug. 2023 - Mar. 2025*
- Sampling and fine-tuning methods for video diffusion models [C1, C3]

# Publications

* denotes equal contribution.

## World Models and Video Generation

[C1] **Temporal In-Context Fine-Tuning for Versatile Control of Video Diffusion Models**
**Kinam Kim***, Junha Hyung*, Jaegul Choo
The Thirty-Ninth Annual Conference on Neural Information Processing Systems (NeurIPS), 2025
**TL;DR:** Temporal In-Context Fine-Tuning (TIC-FT) inserts noisy buffer frames to adapt a pretrained video diffusion model to new conditioning signals, using as few as 10 to 30 samples.

[C2] **EgoX: Egocentric Video Generation from a Single Exocentric Video**
Taewoong Kang*, **Kinam Kim***, Dohyeon Kim*, Minho Park, Junha Hyung, Jaegul Choo
The IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2026
**TL;DR:** EgoX generates egocentric video from a single exocentric video, carrying human appearance and motion through a large viewpoint change while keeping the scene spatially consistent.

[C3] **Spatiotemporal Skip Guidance for Enhanced Video Diffusion Sampling**
Junha Hyung*, **Kinam Kim***, Susung Hong, Min-Jung Kim, Jaegul Choo
The IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2025
*Integrated into LTX-Video and Hugging Face Diffusers*
**TL;DR:** Spatiotemporal Skip Guidance (STG) is a training-free sampling method that raises video diffusion quality by simulating a weak model through selective layer skipping, without the diversity loss of CFG.

[U1] **OmniShowcase: Omnidirectional Subject-Consistent Video Generation from Multi-View Images**
Junyeong Ahn*, Geonwoo Kim*, **Kinam Kim***, Gihwi Kim, Hoonjin Jung, Hyojin Jang, Jaegul Choo
*Under review, 2026*
**TL;DR:** OmniShowcase holds a subject's identity as the camera orbits past surfaces the reference images never showed, by conditioning the video model on surface normals reconstructed from those views. An automated pipeline supplies training videos along full 360-degree orbits, released as Showcase-4K with the ShowcaseBench evaluation set.

[P1] **Cross-Frame Representation Alignment for Fine-Tuning Video Diffusion Models**
Sungwon Hwang*, Hyojin Jang*, **Kinam Kim**, Minho Park, Jaegul Choo
*arXiv preprint, 2025*
**TL;DR:** Cross-frame Representation Alignment (CREPA) regularizes video diffusion fine-tuning against neighboring frames, improving fidelity and cross-frame coherence under LoRA.

[P2] **Overcoming Camera-Facing Gaze Bias in EG3D Scene Generation**
**Kinam Kim**, Juhyun Lee, Youngdo Lee
*arXiv preprint, 2023*
**TL;DR:** Folding a pretrained gaze estimator into EG3D's training pipeline corrects the 3D GAN's camera-facing gaze bias on FFHQ without losing image quality.

## Robot Learning

[C4] **Object-Centric Residual RL for Zero-Shot Sim-to-Real VLA Enhancement**

**Kinam Kim**, Namiko Saito, Heecheol Kim, Katsushi Ikeuchi, Jaegul Choo, Yasuyuki Matsushita

Conference on Robot Learning (CoRL), 2026

*Work done during an internship at Microsoft Research Asia*

**TL;DR:** An object-centric residual policy trained in simulation augments a base VLA and transfers zero-shot to the real world, raising success rates on tasks such as pick-and-place and drawer closing.

[U2] **GistAct: Acting on Latent Answers in Vision-Language-Action Models**

**Kinam Kim**, Fu-En Yang, Min-Hung Chen, Zhiding Yu, Yu-Chiang Frank Wang, Chi-Pin Huang

*Under review, 2026*

*Work done during an internship at NVIDIA*

**TL;DR:** GistAct supervises a VLA's latent reasoning directly rather than distilling it from a text chain-of-thought teacher, ending that reasoning in a goal map over the scene. The action expert reads only the map, never the instruction, and reaches 67.8% success on RoboSPA spatial-24 against 33.9% for the strongest baseline.

[C5] **FlashSAC: Fast and Stable Off-Policy Reinforcement Learning for High-Dimensional Robot Control**

Donghu Kim*, Youngdo Lee*, Minho Park, **Kinam Kim**, I Made Aswin Nahendra, Takuma Seno, Sehee Min, Daniel Palenicek, Florian Vogt, Danica Kragic, Jan Peters, Jaegul Choo, Hojoon Lee

Robotics: Science and Systems (RSS), 2026

*Outstanding Paper Award*

**TL;DR:** FlashSAC cuts Soft Actor-Critic's gradient updates and compensates with larger models and higher throughput, with norm bounds that keep the critic stable. It beats PPO and strong off-policy baselines on over 60 tasks, and trains sim-to-real humanoid locomotion in minutes rather than hours.

[C6] **ACG: Action Coherence Guidance for Flow-based VLA models**

Minho Park*, **Kinam Kim***, Junha Hyung, Hyojin Jang, Hoiyeong Jin, Jooyeol Yun, Hojoon Lee, Jaegul Choo

IEEE International Conference on Robotics and Automation (ICRA), 2026

**TL;DR:** Action Coherence Guidance (ACG) is a training-free test-time method that removes noise-induced jerks and jitter in VLA policies, raising success rates on RoboCasa, DexMimicGen and real-world SO-101.

[U3] **DiscoDemo: Discovering Efficient and Diverse Robot Demonstrations for Imitation Learning**

Minho Park*, **Kinam Kim***, Donghu Kim, Byungkun Lee, Dongyoon Hwang, Yongjae Shin, Junha Hyung, Hojoon Lee, Jaegul Choo

*Under review, 2026*

**TL;DR:** DiscoDemo uses reinforcement learning as a data generator rather than a deployable controller, training a privileged latent-conditioned policy with a reverse curriculum and a success-gated diversity objective to find several distinct solutions per task. The exported demonstrations lift downstream imitation success from 40% to 62% in simulation and from 39% to 78% on a real robot.

[U4] **VLaRL: Augmenting Vision-Language-Action Models with Simulation-Trained Latent-Conditioned Residual RL**

Namiko Saito, **Kinam Kim**, Heecheol Kim, Katsushi Ikeuchi, Yasuyuki Matsushita

*Under review, 2026*

*Work done during an internship at Microsoft Research Asia*

**TL;DR:** VLaRL trains a residual policy for a frozen VLA entirely in simulation and deploys it without any real-world RL, conditioning the residual on the VLA's own vision-language latents and mapping simulation latents toward the real distribution. It improves real-world success on four contact-rich tasks across two VLA backbones.

[U5] **Soft-Value Flow Matching for Offline Reinforcement Learning**

Dohyun Lee, Junhyeong Hong, Theo Choi, **Kinam Kim**, Jaegul Choo, Kyungjae Lee, Yoonseon Oh

*Under review, 2026*

**TL;DR:** Soft-Value Flow Matching (SVFM) improves a flow-matching policy with a learned critic without backpropagating through multi-step sampling, guiding the actor by the gradient of a soft value read off stochastic continuations of a fixed behavior process. It beats the compared baselines on OGBench, LIBERO and LIBERO-Plus, and improves real-world manipulation with both task-specific flow policies and a pretrained VLA.

[O1] **Demonstration-Free Dexterous Manipulation Data at Scale**

**Kinam Kim**, Jaegul Choo

*Ongoing research project, 2026*

**TL;DR:** We generate bimanual dexterous manipulation data without a single human demonstration, on contact-rich tasks such as Rubik's cube turning and bolt fastening. A coding agent composes trajectory optimization, motion planning and scripted control into a task policy, then revises its own code from simulation feedback.

## Skills

- **Languages and frameworks:** Python, PyTorch, multi-GPU and multi-node distributed training
- **Robot learning:** VLA models (GR00T, $\pi_{0.5}$, SmolVLA), flow-matching policies, residual and off-policy RL
- **Simulation and real robots:** MuJoCo, RoboCasa, DexMimicGen, SO-101, sim-to-real transfer
- **Generative models:** video diffusion fine-tuning (Wan2.1, CogVideoX, Mochi), LoRA

## Reference

- **Jaegul Choo**, Ph.D. Advisor, KAIST AI • jchoo@kaist.ac.kr
- **Yu-Chiang Frank Wang**, Internship Manager; Research Director, NVIDIA • ycwang@ntu.edu.tw
- **Yasuyuki Matsushita**, Internship Manager; Senior Director, MSRA Tokyo • yasumat@ist.osaka-u.ac.jp